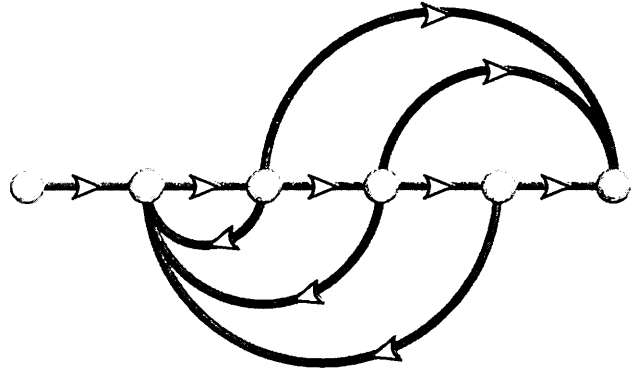


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Time Response

Chapter 4



Chapter Learning Outcomes

After completing this chapter the student will be able to:

- Use poles and zeros of transfer functions to determine the time response of a control system (Sections 4.1–4.2)
- Describe quantitatively the transient response of first-order systems (Section 4.3)
- Write the general response of second-order systems given the pole location (Section 4.4)
- Find the damping ratio and natural frequency of a second-order system (Section 4.5)
- Find the settling time, peak time, percent overshoot, and rise time for an underdamped second-order system (Section 4.6)
- Approximate higher-order systems and systems with zeros as first- or second-order systems (Sections 4.7–4.8)
- Describe the effects of nonlinearities on the system time response (Section 4.9)
- Find the time response from the state-space representation (Sections 4.10–4.11)

Case Study Learning Outcomes

You will be able to demonstrate your knowledge of the chapter objectives with case studies as follows:

- Given the antenna azimuth position control system shown on the front endpapers, you will be able to (1) predict, by inspection, the form of the open-loop angular velocity response of the load to a step voltage input to the power amplifier;

- (2) describe quantitatively the transient response of the open-loop system;
- (3) derive the expression for the open-loop angular velocity output for a step voltage input; (4) obtain the open-loop state-space representation; (5) plot the open-loop velocity step response using a computer simulation.
- Given the block diagram for the Unmanned Free-Swimming Submersible (UFSS) vehicle's pitch control system shown on the back endpapers, you will be able to predict, find, and plot the response of the vehicle dynamics to a step input command. Further, you will be able to evaluate the effect of system zeros and higher-order poles on the response. You also will be able to evaluate the roll response of a ship at sea.

4.1 Introduction

In Chapter 2, we saw how transfer functions can represent linear, time-invariant systems. In Chapter 3, systems were represented directly in the time domain via the state and output equations. After the engineer obtains a mathematical representation of a subsystem, the subsystem is analyzed for its transient and steady-state responses to see if these characteristics yield the desired behavior. This chapter is devoted to the analysis of system transient response.

It may appear more logical to continue with Chapter 5, which covers the modeling of closed-loop systems, rather than to break the modeling sequence with the analysis presented here in Chapter 4. However, the student should not continue too far into system representation without knowing the application for the effort expended. Thus, this chapter demonstrates applications of the system representation by evaluating the transient response from the system model. Logically, this approach is not far from reality, since the engineer may indeed want to evaluate the response of a subsystem prior to inserting it into the closed-loop system.

After describing a valuable analysis and design tool, poles and zeros, we begin analyzing our models to find the step response of first- and second-order systems. The order refers to the order of the equivalent differential equation representing the system—the order of the denominator of the transfer function after cancellation of common factors in the numerator or the number of simultaneous first-order equations required for the state-space representation.

4.2 Poles, Zeros, and System Response

The output response of a system is the sum of two responses: the *forced response* and the *natural response*.¹ Although many techniques, such as solving a differential equation or taking the inverse Laplace transform, enable us to evaluate this output response, these techniques are laborious and time-consuming. Productivity is aided by analysis and design techniques that yield results in a minimum of time. If the technique is so rapid that we feel we derive the desired result by inspection, we sometimes use the attribute *qualitative* to describe the method. The use of poles and

¹ The forced response is also called the *steady-state response* or *particular solution*. The natural response is also called the *homogeneous solution*.

This section defined two specifications, or parameters, of second-order systems: natural frequency, ω_n , and damping ratio, ζ . We saw that the nature of the response obtained was related to the value of ζ . Variations of damping ratio alone yield the complete range of overdamped, critically damped, underdamped, and undamped responses.

4.6 Underdamped Second-Order Systems

Now that we have generalized the second-order transfer function in terms of ζ and ω_n , let us analyze the step response of an *underdamped* second-order system. Not only will this response be found in terms of ζ and ω_n , but more specifications indigenous to the underdamped case will be defined. The underdamped second-order system, a common model for physical problems, displays unique behavior that must be itemized; a detailed description of the underdamped response is necessary for both analysis and design. Our first objective is to define transient specifications associated with underdamped responses. Next we relate these specifications to the pole location, drawing an association between pole location and the form of the underdamped second-order response. Finally, we tie the pole location to system parameters, thus closing the loop: Desired response generates required system components.

Let us begin by finding the step response for the general second-order system of Eq. (4.22). The transform of the response, $C(s)$, is the transform of the input times the transfer function, or

$$C(s) = \frac{\omega_n^2}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)} = \frac{K_1}{s} + \frac{K_2 s + K_3}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (4.26)$$

where it is assumed that $\zeta < 1$ (the underdamped case). Expanding by partial fractions, using the methods described in Section 2.2, Case 3, yields

$$C(s) = \frac{1}{s} - \frac{(s + \zeta\omega_n) + \frac{\zeta}{\sqrt{1-\zeta^2}}\omega_n\sqrt{1-\zeta^2}}{(s + \zeta\omega_n)^2 + \omega_n^2(1-\zeta^2)} \quad (4.27)$$

Taking the inverse Laplace transform, which is left as an exercise for the student, produces

$$\begin{aligned} c(t) &= 1 - e^{-\zeta\omega_n t} \left(\cos \omega_n \sqrt{1-\zeta^2} t + \frac{\zeta}{\sqrt{1-\zeta^2}} \sin \omega_n \sqrt{1-\zeta^2} t \right) \\ &= 1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \cos(\omega_n \sqrt{1-\zeta^2} t - \phi) \end{aligned} \quad (4.28)$$

where $\phi = \tan^{-1}(\zeta/\sqrt{1-\zeta^2})$.

A plot of this response appears in Figure 4.13 for various values of ζ , plotted along a time axis normalized to the natural frequency. We now see the relationship between the value of ζ and the type of response obtained: The lower the value of ζ , the more oscillatory the response. The natural frequency is a time-axis scale factor and does not affect the nature of the response other than to scale it in time.

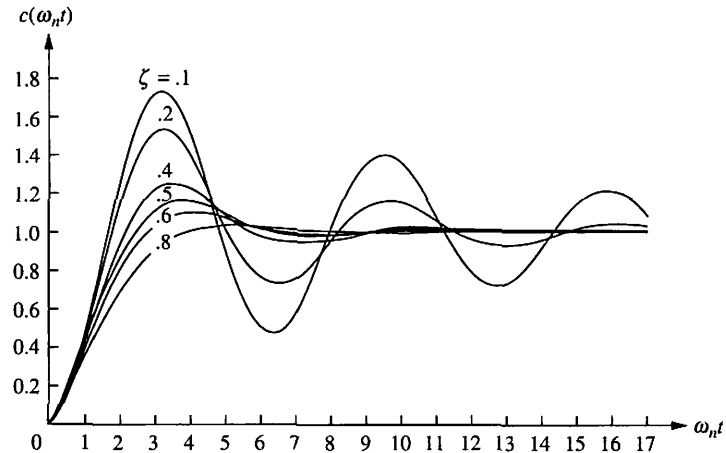


FIGURE 4.13 Second-order underdamped responses for damping ratio values

We have defined two parameters associated with second-order systems, ζ and ω_n . Other parameters associated with the underdamped response are rise time, peak time, percent overshoot, and settling time. These specifications are defined as follows (see also Figure 4.14):

1. *Rise time, T_r .* The time required for the waveform to go from 0.1 of the final value to 0.9 of the final value.
2. *Peak time, T_p .* The time required to reach the first, or maximum, peak.
3. *Percent overshoot, %OS.* The amount that the waveform overshoots the steady-state, or final, value at the peak time, expressed as a percentage of the steady-state value.
4. *Settling time, T_s .* The time required for the transient's damped oscillations to reach and stay within $\pm 2\%$ of the steady-state value.

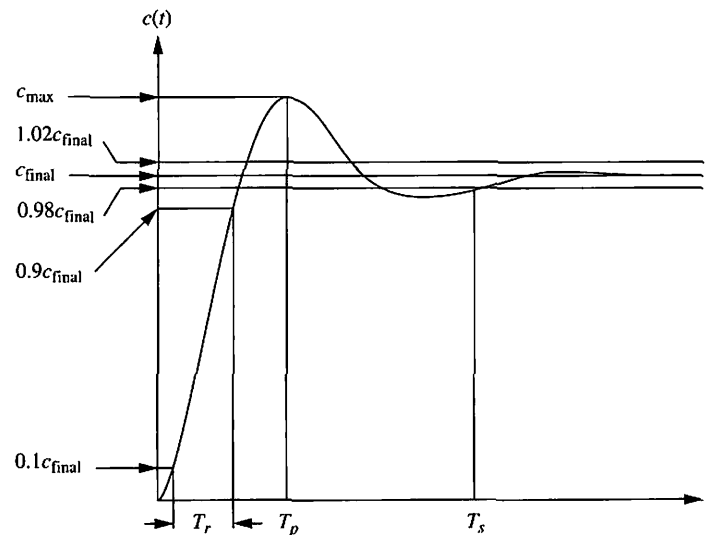


FIGURE 4.14 Second-order underdamped response specifications

Notice that the definitions for settling time and rise time are basically the same as the definitions for the first-order response. All definitions are also valid for systems of order higher than 2, although analytical expressions for these parameters cannot be found unless the response of the higher-order system can be approximated as a second-order system, which we do in Sections 4.7 and 4.8.

Rise time, peak time, and settling time yield information about the speed of the transient response. This information can help a designer determine if the speed and the nature of the response do or do not degrade the performance of the system. For example, the speed of an entire computer system depends on the time it takes for a hard drive head to reach steady state and read data; passenger comfort depends in part on the suspension system of a car and the number of oscillations it goes through after hitting a bump.

We now evaluate T_p , %OS, and T_s as functions of ζ and ω_n . Later in this chapter we relate these specifications to the location of the system poles. A precise analytical expression for rise time cannot be obtained; thus, we present a plot and a table showing the relationship between ζ and rise time.

Evaluation of T_p

T_p is found by differentiating $c(t)$ in Eq. (4.28) and finding the first zero crossing after $t = 0$. This task is simplified by “differentiating” in the frequency domain by using Item 7 of Table 2.2. Assuming zero initial conditions and using Eq. (4.26), we get

$$\mathcal{L}[\dot{c}(t)] = sC(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (4.29)$$

Completing squares in the denominator, we have

$$\mathcal{L}[\dot{c}(t)] = \frac{\omega_n^2}{(s + \zeta\omega_n)^2 + \omega_n^2(1 - \zeta^2)} = \frac{\frac{\omega_n}{\sqrt{1 - \zeta^2}} \omega_n \sqrt{1 - \zeta^2}}{(s + \zeta\omega_n)^2 + \omega_n^2(1 - \zeta^2)} \quad (4.30)$$

Therefore,

$$\dot{c}(t) = \frac{\omega_n}{\sqrt{1 - \zeta^2}} e^{-\zeta\omega_n t} \sin \omega_n \sqrt{1 - \zeta^2} t \quad (4.31)$$

Setting the derivative equal to zero yields

$$\omega_n \sqrt{1 - \zeta^2} t = n\pi \quad (4.32)$$

or

$$t = \frac{n\pi}{\omega_n \sqrt{1 - \zeta^2}} \quad (4.33)$$

Each value of n yields the time for local maxima or minima. Letting $n = 0$ yields $t = 0$, the first point on the curve in Figure 4.14 that has zero slope. The first peak, which occurs at the peak time, T_p , is found by letting $n = 1$ in Eq. (4.33):

$$T_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}} \quad (4.34)$$

Evaluation of %OS

From Figure 4.14 the percent overshoot, %OS, is given by

$$\%OS = \frac{c_{\max} - c_{\text{final}}}{c_{\text{final}}} \times 100 \quad (4.35)$$

The term $c_{\max}$ is found by evaluating $c(t)$ at the peak time, $c(T_p)$. Using Eq. (4.34) for T_p and substituting into Eq. (4.28) yields

$$\begin{aligned} c_{\max} = c(T_p) &= 1 - e^{-(\zeta\pi/\sqrt{1-\zeta^2})} \left(\cos \pi + \frac{\zeta}{\sqrt{1-\zeta^2}} \sin \pi \right) \\ &= 1 + e^{-(\zeta\pi/\sqrt{1-\zeta^2})} \end{aligned} \quad (4.36)$$

For the unit step used for Eq. (4.28),

$$c_{\text{final}} = 1 \quad (4.37)$$

Substituting Eqs. (4.36) and (4.37) into Eq. (4.35), we finally obtain

$$\%OS = e^{-(\zeta\pi/\sqrt{1-\zeta^2})} \times 100 \quad (4.38)$$

Notice that the percent overshoot is a function only of the damping ratio, ζ .

Whereas Eq. (4.38) allows one to find %OS given ζ , the inverse of the equation allows one to solve for ζ given %OS. The inverse is given by

$$\zeta = \frac{-\ln(\%OS/100)}{\sqrt{\pi^2 + \ln^2(\%OS/100)}} \quad (4.39)$$

The derivation of Eq. (4.39) is left as an exercise for the student. Equation (4.38) (or, equivalently, (4.39)) is plotted in Figure 4.15.

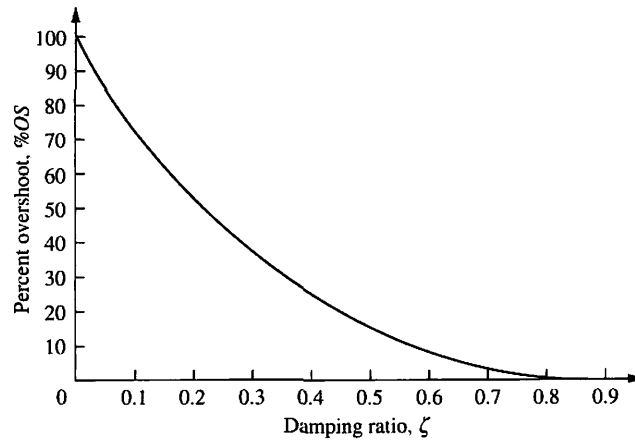


FIGURE 4.15 Percent overshoot versus damping ratio

Evaluation of T_s

In order to find the settling time, we must find the time for which $c(t)$ in Eq. (4.28) reaches and stays within $\pm 2\%$ of the steady-state value, c_{final} . Using our definition, the settling time is the time it takes for the amplitude of the decaying sinusoid in Eq. (4.28) to reach 0.02, or

$$e^{-\zeta\omega_n t} \frac{1}{\sqrt{1-\zeta^2}} = 0.02 \quad (4.40)$$

This equation is a conservative estimate, since we are assuming that $\cos(\omega_n \sqrt{1-\zeta^2}t - \phi) = 1$ at the settling time. Solving Eq. (4.40) for t , the settling time is

$$T_s = \frac{-\ln(0.02\sqrt{1-\zeta^2})}{\zeta\omega_n} \quad (4.41)$$

You can verify that the numerator of Eq. (4.41) varies from 3.91 to 4.74 as ζ varies from 0 to 0.9. Let us agree on an approximation for the settling time that will be used for all values of ζ ; let it be

$$T_s = \frac{4}{\zeta\omega_n} \quad (4.42)$$

Evaluation of T_r

A precise analytical relationship between rise time and damping ratio, ζ , cannot be found. However, using a computer and Eq. (4.28), the rise time can be found. We first designate $\omega_n t$ as the normalized time variable and select a value for ζ . Using the computer, we solve for the values of $\omega_n t$ that yield $c(t) = 0.9$ and $c(t) = 0.1$. Subtracting the two values of $\omega_n t$ yields the normalized rise time, $\omega_n T_r$, for that value of ζ . Continuing in like fashion with other values of ζ , we obtain the results plotted in Figure 4.16.⁵ Let us look at an example.

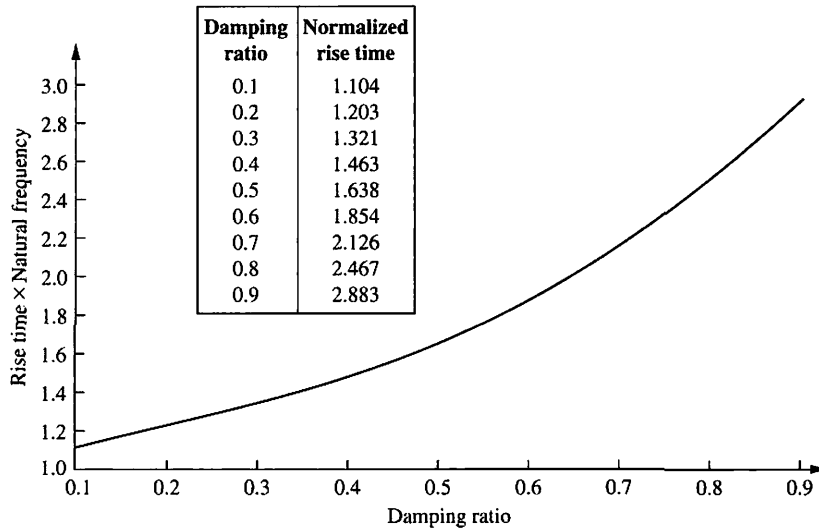


FIGURE 4.16 Normalized rise time versus damping ratio for a second-order underdamped response

⁵ Figure 4.16 can be approximated by the following polynomials: $\omega_n T_r = 1.76\zeta^3 - 0.417\zeta^2 + 1.039\zeta + 1$ (maximum error less than $\frac{1}{2}\%$ for $0 < \zeta < 0.9$), and $\zeta = 0.115(\omega_n T_r)^3 - 0.883(\omega_n T_r)^2 + 2.504(\omega_n T_r) - 1.738$ (maximum error less than 5% for $0.1 < \zeta < 0.9$). The polynomials were obtained using MATLAB's **polyfit** function.